

Methodology and rationale

This section details the hydrological basis of the tool, explaining the selection of parameters, their physical significance, and underlying assumptions.

To calculate geomorphic FSI (flood susceptibility index) six parameters are used. Each parameter is computed at the subbasin scale. How they are found and their significance is given below.

1) **Drainage density**: It is the length of streams/channels per unit area (km/km^2), and it measures the efficiency of runoff conveyance. A higher drainage density allows faster routing.

How it's calculated: Stream vectors are intersected with the subbasin polygons to find total stream length per subbasin, which is divided by subbasin area to arrive to drainage density.

The stream network density is controlled by the threshold for any watershed. To ensure uniformity between analyses, the streams used for drainage density and HAND are streams with a contributing area of 5 km^2 , which corresponds to naturally occurring stream initiation. It is for this reason that this tool is designed for use with a 30m DEM because streams are extracted at a threshold of 5500 (5km^2) for the purposes of drainage density and HAND (given below).

2) **HAND** (Height Above Nearest Drainage): It is the vertical elevation of a point to the nearest stream channel. Lower values of HAND indicate that the area is more susceptible to inundation.

How it's calculated: Using `r.stream.distance` from GRASS, a raster of HAND is produced, using streams with a threshold of 5500. Since HAND is a comparable metric in any watershed, a value of 5m is used as a threshold. Areas where $\text{HAND} < 5\text{m}$ are found using raster calculations and for each subbasin a fraction is computed:

$$\text{HAND}_{\text{frac}} = \frac{\text{Area where HAND} < 5\text{m}}{\text{Subbasin area}}$$

A higher fraction indicates a subbasin has a higher fraction of total area susceptible to inundation.

3) **TWI** (Topographic Wetness Index): TWI is a proxy for soil moisture and potential saturation, derived from upslope contributing area and local slope.

How it's calculated: It is calculated using the built-in output option (Topographic Index) in r.watershed. To capture spatial connectivity and to reduce noise in the data, only areas where TWI is high are considered. This is done by finding the 90th percentile of the TWI value in the watershed, termed TWI_{crit}, and considering areas where TWI > TWI_{crit}. Similar to HAND a fraction is calculated for each subbasin.

$$TWI_{exceedance_frac} = \frac{\text{Area where TWI} > \text{TWI}_{crit}}{\text{Subbasin area}}$$

4) **Soil transmissivity:** It is a measure of how easily water flows through soils and is derived through hydraulic conductivity (K_{sat}) and soil depth. Soil transmissivity's effect on flooding is timescale dependent.

High soil transmissivity promotes infiltration and reduces short-term (flash) runoff. During prolonged rainfall it leads to more baseflow recharge for rivers potentially increasing river levels.

How it's calculated: Global rasters for soil depth and hydraulic conductivity with the same resolution (1km²) are overlain and clipped to the watershed boundary. Units are made consistent and both rasters are multiplied to obtain soil transmissivity in m/day.

5) **Basin shape factor (Kc):** Also termed as the compactness coefficient, it describes how circular or elongated a basin is. Circular basins have a value closer to 1 and are associated with synchronised runoff, higher peak discharges and are hence more flood prone. On the other hand, elongated basins have values higher than 1, have delayed runoff and lower peaks which lead to a reduced flood intensity.

How it's calculated: Using QGIS' feature geometry options each basin polygon's Kc can be found using the formula below.

$$Kc = \frac{\text{Basin perimeter}}{2\sqrt{\pi * \text{Basin area}}}$$

6) **Runoff coefficient (C):** The fraction of rainfall that becomes surface runoff. High runoff coefficients correspond to areas of lower infiltration and hence increased flood susceptibility.

How it's calculated: The SCS curve method is employed, and it is a multistep calculation as detailed below.

I) ESA world-cover rasters (at 30m resolution) are clipped to the watershed area to find the landcover type.

II) The rasters are reclassified based on the landcover type to indicate an approximate CN number. The reclassification is given below.

Landcover Type	CN
Treecover	55
Shrubland	65
Grassland	70
Cropland	80
Built-up	90
Bare/sparse vegetation	85
Snow and ice	95
Permanent water bodies	100
Herbaceous wetlands	75
Mangroves	70
Moss and lichen	60

III) To improve reliability of these CN values they are adjusted based on local slope and antecedent moisture condition. R.slope.aspect is used to find the slope raster. The annual rainfall raster, generated by the rainfall script in Miniconda, is used to approximate antecedent moisture condition (AMC). The adjustments for both slope and AMC are given below.

Slope (%)	Adjustment to CN
$5 < \text{slope} \leq 10$	+2
$10 < \text{slope} \leq 20$	+5
$20 < \text{slope}$	+8

AMC (indicated by annual rainfall in mm)	Adjustment	Description
Annual rainfall < 350 (dry conditions)	$\frac{CN}{2.281 - 0.01281CN}$	Reduction in CN to account for more infiltration capacity
$350 < \text{Annual rainfall} \leq 550$	CN	No adjustment
$550 < \text{Annual rainfall}$ (wet conditions)	$\frac{CN}{0.427 + 0.00573CN}$	Increase in CN to account for more saturated soils

IV) Calculate runoff depth Q from the equation

$$Q = \frac{(P - 0.2S)^2}{(P + 0.8S)}$$

Where P is the design storm value which is the 95th percentile of daily rainfall based on the CHIRPS data set (see Rainfall below) and

$$S = \frac{25400}{CN} - 254$$

Finally, the runoff coefficient (C) is calculated as

$$C = \frac{Q}{P} \text{ for } P > 0.2S$$

Those were the six geomorphic parameters that are used in this tool to indicate how terrain and geology affect flooding. However, two more factors are found to paint a more realistic picture of flood risk in the watershed: rainfall and exposure.

Rainfall

Rainfall acts as the primary forcing mechanism in the model. CHIRPS global NetCDF data is used to derive rainfall statistics.

Processing

Using Xarray (Python library), the precipitation data is loaded and clipped to the watershed boundary. Days with precipitation ≤ 0 are excluded ('wet days' only). For each 1 km² grid cell within the watershed, the 95th percentile of rainfall over all wet days in the time series is computed. This value serves as the design storm depth for that cell. Additionally, the data is resampled to the year end and the mean annual rainfall raster is also created for the antecedent moisture conditions approximation. All data is then resampled to 30m resolution to match the DEM.

Exposure

Exposure represents what is at risk during a flood event and reflects potential socio-economic impact.

The tool uses ESA landcover rasters (10m resolution) clipped to the watershed to understand the type and extent of landcover. Exposure values are assigned to the type of landcover as shown in the table below.

Landcover Type	Risk (0–1)
Treecover	0.2
Shrubland	0.2
Grassland	0.3
Cropland	0.7
Built-up	1.0
Bare/sparse vegetation	0.0
Snow and ice	0.0
Permanent water bodies	0.0
Herbaceous wetlands	0.1
Mangroves	0.1
Moss and lichen	0.1

For each subbasin in the watershed the weighted mean of exposure is calculated to indicate at how much risk the watershed is.

Final calculations

This section explains the final equations used to combine the factors into single meaningful values over the watershed area which are:

Geomorphic Flood Susceptibility Index (FSI_geomorph)
 Rainfall Adjusted Flood Susceptibility Index (FSI_rainfall)
 Flood Risk
 Percentage Hillslope Contribution

Since all factors have different units (m/day, km/km, dimensionless etc.) they are normalised to a scale of 0-1 before they are used in calculations.

First, two linear combinations of the factors are found.

Runoff Generating Potential (RGP)

$$RGP = 0.3(TWL_{exceedance}) + 0.35(Runoff_{coefficient}) + 0.15 * (Slope) + 0.2 * (1 - Transmissivity)$$

Routing and Floodplain Potential (RFP)

$$RFP = 0.35(Drainage_{density}) + 0.35(HAND_{frac}) + 0.1 * (Transmissivity) + 0.2(1 - Shapefactor_{Kc})$$

The RGP equation corresponds to a hillslope regime flood. This can be explained by looking at the factors and their weighting.

High TWI: More saturated soils, greater runoff potential

High Runoff Coefficient: Greater runoff

High Slope: Faster downslope movement of runoff, less infiltration

Low Transmissivity: Less groundwater recharge, thinner more impermeable soils, greater runoff

Similarly, the RFP equations variables correlate to riverine flooding

High Drainage Density: Greater number of channels to direct runoff, increased river levels

Low HAND (Height above nearest drainage): Indicated by a high *HAND_frac*, it relates the fraction of the subbasin area susceptible to rising river levels.

High Transmissivity: Greater baseflow, increasing river levels.

Low Shape factor: More circular subbasins, more synchronized flow, higher peaks

Then a dimensionless weighting factor which shows how riverine a particular subbasin is found as

$$Riverine\ index\ (RI) = \frac{RFP}{(RFP + RGP)}$$

Finally, Geomorphic Flood Susceptibility is found as a weighted blend of RFP and RGP using the Riverine index

$$FSI_{geomorph} = RI(RFP) + (1 - RI)(RGP)$$

And the percentage hillslope contribution to FSI is found as

$$\% \text{ Hillslope Contribution} = \frac{(1 - RI)(RGP)}{FSI_{geomorph}}$$

It is found that due to spatial smoothing and the infrared based retrieval methods of CHIRPS rainfall data, extreme rainfall events are often underestimated. This particularly affects the hillslope flooding regime as it is triggered by short term high rainfall. To correct this bias, an empirical factor of 1.25 for rainfall in the hillslope regime is introduced.

Rainfall Adjusted Flood Susceptibility Index ($FSI_{rainfall}$) is calculated as

$$FSI_{rainfall} = \overset{\text{Riverine}}{[RI * RFP * P95_n]} + \overset{\text{Hillslope}}{[1.25 * (1 - RI) * RGP * P95_n]}$$

Where $P95_n$ is the normalised extreme rainfall value of the subbasin within the watershed.

The final most relevant metric, Flood Risk, is calculated by combining $FSI_{rainfall}$ with the exposure of the basin.

$$Flood\ Risk = FSI_{rainfall} * Exposure$$

Multicollinearity

Variance Inflation Factor (VIF) was evaluated to assess multicollinearity among the six geomorphic parameters in the model. Multicollinearity occurs when explanatory variables are highly correlated, which can distort the relative importance of each factor and lead to unstable or misleading model weights.

By computing VIF values for the parameters for multiple watershed locations it was found that they were all below 5 indicating each parameter contributed independent information to the model.

Limitations

As with many models, inaccuracies and limitations exist and here are the most relevant ones of this tool.

- This model is a climate-based model and is static in nature. It does not account for drastic changes in weather or highly localised events but rather a historic overview of conditions. This is seen in the approximation of the antecedent soil moisture condition which uses annual precipitation amounts as a proxy, when in practice the sum of precipitation over the past five days is what is used.
- There is no consideration of human intervention in flood prevention such as dikes, dams, channel engineering, flood walls etc. These are commonly employed, effective strategies to mitigate impact and are left out of the analysis due to the global scale of this model. Procuring worldwide flood preventative measures would be difficult and hard to standardize.
- The final Flood Risk map, or FSI map shows relative risk or susceptibility within the watershed, which means that there is no direct quantitative comparison between a subbasin in Italy and Japan, and a low risk subbasin may in fact be of significance. A different model (standardized and global) will be needed to capture this.
- Complex events that are pertinent to flooding are not captured: for example, rain on snow events, back-to-back storms, ice jam floods, infrastructure failure such as dam or levee breaches etc.
- The landcover types from the ESA WorldCover rasters provide only an approximation to CN values which are used for infiltration. Dominant landcover types such as treecover, grasslands and croplands take on one CN value and have no in-class variation which would be observed in real life.